ModelIt!

Teacher's Guide

What Defines the Ethics of Science?

Modeling Scientific Responsibility, Dual-Use Research, and Evidence-Based Policy

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ETS1-3

Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate competing ethical claims about dual-use research, scientific freedom, and public benefit, constructing evidence-based arguments about governance frameworks using model-derived data about the relationships between oversight, innovation, and societal impact.
Disciplinary Core Idea	**ETS1.B Developing Possible Solutions** — Students evaluate solutions to the complex problem of balancing scientific innovation with ethical responsibility, considering constraints including dual-use risk, equity, public trust, and international governance coordination.
Crosscutting Concept	**Cause and Effect** — Students trace the causal mechanisms through which funding decisions, oversight structures, and publication norms shape the trajectory of scientific research and determine whether its benefits and risks are equitably distributed.

CAST Testing Alignment

MULTIPLE CHOICE:

A research institution publishes detailed methods for engineering a highly transmissible pathogen, arguing that transparency enables the scientific community to develop countermeasures. Based on the science ethics model, what is the predicted effect on Societal Impact Score and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a scientific system with maximum Innovation Rate but low Institutional Oversight Rigor and poor Benefit Distribution Equity would produce a lower Societal Impact Score than a system with moderate Innovation Rate, strong oversight, and equitable distribution. Reference at least three model components and their relationships.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Science is not a neutral tool — it is a human activity shaped by values, funding, politics, and power. The same nuclear physics that generates clean electricity also created weapons capable of ending civilization. The same genetic engineering that produces life-saving medicines also enables biological weapons. The same artificial intelligence that diagnoses cancer also powers autonomous weapons and mass surveillance. As scientific power increases exponentially, the ethical frameworks governing its use have struggled to keep pace.

The Social Contract of Science

Since World War II, science has operated under an implicit social contract: society funds basic research through taxes, and in return, science delivers innovations that improve health, prosperity, and security. This contract depends on public trust — the belief that scientific institutions act honestly and in the public interest. When trust erodes (through perceived conflicts of interest, failure to address community concerns, or high-profile fraud), public support for science funding declines, vaccine hesitancy increases, and evidence-based policy becomes politically impossible. Rebuilding trust requires transparency, accountability, community engagement, and honest acknowledgment of uncertainty and limitations.

The Dual-Use Dilemma

Dual-use research of concern (DURC) creates knowledge that can both benefit and threaten humanity. The most dramatic example is gain-of-function virology: researchers deliberately make pathogens more transmissible or virulent to understand pandemic mechanisms and develop countermeasures. This research helped prepare for COVID-19 but also creates blueprints for potential bioweapons. In 2011, researchers created a highly transmissible H5N1 influenza strain and sought to publish the methods. The resulting controversy led to a US government moratorium on gain-of-function research from 2014-2017 and the development of new oversight frameworks. The core dilemma: restricting publication prevents misuse but also prevents the scientific scrutiny that catches errors, and blocking research in one country simply moves it to less regulated jurisdictions.

Equity in Science

Scientific benefits and risks are not distributed equally. Clinical trials historically over-enrolled white male participants, producing treatments optimized for one demographic. Environmental research facilities and hazardous waste sites are disproportionately located in low-income communities and communities of color. Genomic databases predominantly represent European ancestry, limiting the precision medicine benefits available to other populations. The COVID-19 pandemic revealed that vaccine distribution tracked national wealth, with high-income countries achieving 70% coverage while low-income countries remained below 20%. Equitable science requires deliberate design: representative clinical trials, community-engaged research, open-access publication, technology transfer to developing nations, and governance structures that give affected communities voice in research decisions.

Emerging Ethical Frontiers

The 21st century presents ethical challenges that existing frameworks were not designed to address. AI systems that make life-altering decisions (criminal sentencing, hiring, medical diagnosis) encode biases from training data and operate with limited transparency. Heritable human gene editing (germline CRISPR) could eliminate genetic diseases but also enable genetic enhancement, potentially creating genetic inequity between those who can and cannot afford modification. Solar geoengineering (stratospheric aerosol injection) could cool the planet but would affect weather patterns globally, raising questions about who has the right to modify Earth's climate. Synthetic biology enables engineering entirely new organisms, with applications from biofuels to biological weapons. Each frontier requires new ethical frameworks developed through inclusive dialogue between scientists, ethicists, policymakers, and the communities affected by the research.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the science ethics system: funding allocation, dual-use potential, public trust, institutional oversight, publication transparency, researcher ethics, benefit equity, risk equity, innovation rate, and societal impact.

E **Establish Relationships**

Students map relationships showing how oversight and transparency build public trust, how trust supports funding and collaboration, how dual-use potential creates tension between openness and restriction, and how equity in benefits and risks determines whether scientific progress serves all of society.

V **Visualize & Model**

Students build a ten-component computational model showing the interacting dynamics of scientific governance, with feedback loops between trust, transparency, funding, and innovation.

E **Evaluate Outcomes**

Students run standard research, dual-use crisis, and trust collapse scenarios to quantify how ethical governance structures affect the net societal impact of scientific research.

R **Revise & Extend**

Students add international governance coordination, intellectual property frameworks, or whistleblower protection to model the institutional and legal dimensions of scientific ethics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Science gave us vaccines that save millions of lives and nuclear weapons that could end civilization. Gene editing that cures disease and tools for genetic discrimination. AI that diagnoses cancer and autonomous weapons that kill without human decision. The same knowledge does both. Who decides which applications move forward?"*

Watch For

Show cover image with contrasting dual-use examples. Ask students: can you think of a scientific technology that has both saved and destroyed lives? Build a class list of dual-use examples.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling something most science classes never touch: the ethics system that governs scientific research. You will discover that ethics is not a brake on science but a steering wheel — it determines whether scientific power serves humanity or harms it."*

Watch For

Have students read objectives. Pre-teach 'dual-use' with a clear example: the same chemistry that produces fertilizer to grow food also produces explosives for weapons (the Haber-Bosch process). Ask: should Fritz Haber have been stopped?

Slide 3: Big Question

3 min

Say: *"Who decides which research gets done? Who benefits? Who bears the risks? And what happens when the same knowledge that could save millions could also kill millions? These are not philosophical abstractions — they are urgent questions that scientists, policymakers, and citizens must answer NOW."*

Watch For

Quick-write: Students write one scientific technology they believe should have stronger ethical oversight and explain why. Share out and discuss: what criteria should determine the level of oversight?

Slide 4: LEVER Framework

2 min

Say: *"We will Label the components that determine whether scientific research produces net benefit, Establish the feedback loops between trust, transparency, and innovation, Visualize the ethics system model, Evaluate three governance scenarios, and Refine by adding international coordination and whistleblower protection."*

Watch For

Preview each LEVER step. Emphasize that this is the capstone lesson because all the science they have studied this year operates within an ethical system — understanding that system is as important as understanding the science itself.

Slide 5: Activity 1: Components

8 min

Say: *"Three inputs set the context: funding determines what gets studied, dual-use potential determines the risk level, and public trust determines whether society supports science. Seven internal components represent the governance mechanisms that determine whether research produces net benefit or net harm."*

Watch For

Students sort external versus internal components. Debate: is Public Trust truly external or can scientists influence it? This leads to discussing the feedback loop between transparency, trust, and funding.

Slide 6: Activity 2: Connections

8 min

Say: *"Transparency builds trust. Trust supports funding. Funding enables innovation. But dual-use potential creates tension: maximum transparency enables scrutiny AND misuse. Maximum restriction prevents misuse AND progress. Find the feedback loops and identify the tensions."*

Watch For

Students draw arrows showing relationships. Identify the trust-transparency positive feedback loop and the dual-use tension between openness and restriction. Discuss: how do you optimize a system with inherent trade-offs?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three scenarios: standard research with low ethical risk, a dual-use crisis with extreme misuse potential, and a public trust collapse. Watch how the Societal Impact Score responds to each scenario and identify which governance mechanisms matter most."*

Watch For

Students run all three scenarios and compare outcomes. Key discussion: the trust collapse scenario shows cascading effects across ALL components. Ask: what does this tell us about the importance of maintaining public trust even when it is inconvenient?

Slide 8: Discoveries

5 min

Say: *"Your model reveals that ethics is not a constraint on science but a system property that determines its value. Maximum innovation without governance produces lower societal impact than moderate innovation with strong governance. Public trust is the foundation everything else depends on. And equity is not optional — it is an ethical and practical necessity."*

Watch For

Class discussion of discoveries. This is the capstone moment: connect this lesson to every previous lesson. Every system they modeled this year — from Mars colonies to cancer treatment to fusion power — exists within an ethical governance system. Ask: how does understanding ethics change how you think about every other science topic?

Slide 9: STEM Challenge

12 min

Say: *"You are designing the ethical governance framework for a $5 billion research institute pursuing AI, synthetic biology, and climate engineering. These are the most powerful and dangerous technologies in human history. Your framework must enable breakthrough innovation while preventing catastrophic misuse. This is the hardest design challenge in the entire curriculum — because the stakes are the highest."*

Watch For

Groups design governance frameworks with ethics review structures, dual-use classification systems, publication policies, equity mandates, and public engagement mechanisms. Present and evaluate: which framework best balances innovation and responsibility? Which is most equitable? Which would you trust with the most powerful technologies?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Research Funding Allocation, Dual-Use Potential, Public Trust in Science
- **INTERNAL:** Institutional Oversight Rigor, Publication Transparency, Researcher Ethical Awareness, Benefit Distribution Equity, Risk Distribution Equity, Innovation Rate, Societal Impact Score

Relationships

- **Research Funding Allocation to Innovation Rate:** POSITIVE (+) — More funding enables more researchers, more experiments, and more rapid progress. However, the relationship is not linear — funding directed by short-term political priorities may not produce the most impactful innovation.
- **Institutional Oversight Rigor to Innovation Rate:** NEGATIVE (-) — Stronger oversight creates additional requirements that slow the research process — ethics reviews, safety protocols, and approval processes take time. However, this slight reduction in speed prevents catastrophic harms that would cause much larger setbacks to innovation through scandals, moratoriums, and trust loss.
- **Publication Transparency to Public Trust in Science:** POSITIVE (+) — Transparent sharing of methods, data, and results — including negative results and honest acknowledgment of uncertainty — builds public confidence in the integrity and accountability of scientific institutions.
- **Public Trust in Science to Research Funding Allocation:** POSITIVE (+) — Higher public trust generates greater political will to fund scientific research through tax-supported agencies. When trust declines, politicians face pressure to cut research budgets and impose restrictions.
- **Dual-Use Potential to Institutional Oversight Rigor:** POSITIVE (+) — Research with higher dual-use potential triggers more intensive ethical review, biosafety requirements, and publication restrictions. The governance system should scale oversight proportionally to risk.
- **Benefit Distribution Equity to Societal Impact Score:** POSITIVE (+) — When scientific benefits are equitably distributed across communities and nations, the total societal impact is higher because more people benefit. Concentrated benefits produce lower overall impact and generate resentment that erodes public trust.
- **Risk Distribution Equity to Societal Impact Score:** POSITIVE (+) — When research risks are equitably distributed rather than concentrated on marginalized communities, societal impact improves because no group bears disproportionate harm. Equitable risk distribution also maintains trust and support from all communities.
- **Researcher Ethical Awareness to Publication Transparency:** POSITIVE (+) — Scientists with strong ethical training and awareness are more likely to practice transparent reporting, disclose conflicts of interest, share data openly, and report negative results honestly — all of which strengthen the self-correcting nature of science.

Key Discoveries

- Scientific ethics is not a constraint on science but a system property that determines whether science produces net benefit or net harm to society — the model reveals that maximum Innovation Rate does not maximize Societal Impact Score because unregulated innovation produces inequitable distribution, public trust erosion, and dual-use risks that reduce overall benefit
- Publication transparency and institutional oversight create a positive feedback loop with public trust — transparent, well-regulated science builds trust, which supports funding and collaboration, which accelerates responsible innovation, while secrecy and misconduct erode trust in a self-reinforcing downward spiral
- Benefit and risk distribution equity are independent variables that must be separately optimized — a scientific advance can have enormous total benefit but negative Societal Impact Score if benefits flow only to wealthy communities while risks burden marginalized populations

- The dual-use dilemma has no universal solution — the model reveals that the optimal balance between scientific freedom and restriction depends on the specific dual-use potential of the research, the availability of defensive applications, and the governance capacity to prevent misuse, making case-by-case ethical evaluation essential rather than blanket rules

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ethics slows down science and reduces innovation"	While ethics review adds process time, the overall impact on innovation is positive. Ethical research is more reliable (because it is conducted carefully and transparently), more trusted (because the public supports it), better funded (because trust enables political support), and less likely to produce catastrophic setbacks (because risks are managed proactively). The history of science shows that ethical failures — Tuskegee, thalidomide, the He Jiankui gene-editing scandal — cause moratoriums, funding cuts, and loss of public trust that set entire fields back by years or decades. Ethics review is an investment in sustainable innovation, not a cost.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Scientists are objective and do not need external ethical oversight"	Scientists are human beings subject to the same biases, incentive structures, and conflicts of interest as everyone else. The academic incentive system rewards publication quantity, novelty, and positive results — creating pressure to cut corners, overstate findings, or ignore ethical concerns that might delay publication. External oversight provides independent checks that complement individual integrity. Every profession with significant power over others — medicine, law, engineering, finance — has external regulatory oversight. Science should not be exempt simply because scientists believe they are objective.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The benefits of science automatically reach everyone equally"	Scientific benefits follow existing power and wealth structures unless deliberately redirected. Pharmaceutical companies develop drugs for profitable markets (wealthy nations) rather than unprofitable ones (malaria, tuberculosis). Precision medicine advances fastest for populations overrepresented in genomic databases (European ancestry). Clean energy technology is deployed first in nations that can afford it. COVID-19 vaccines reached wealthy countries months before poor countries despite the virus not respecting borders. Equitable distribution requires intentional design: tiered pricing, technology transfer agreements, representative clinical trials, and community-engaged research priorities.